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Monitoring IMS Synchronous Change Data Capture

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Abstract

This article describes how to use the PowerExchange *x*EDP-STAT and *x*EDP-STATWTO external subsystem commands to monitor PowerExchange change capture activity for IMS synchronous CDC.

Supported Versions

- PowerExchange 10.0 or later

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Overview

To determine if IMS synchronous change capture is active and the level of capture activity, issue the PowerExchange *x*EDP_STAT or *x*EDP-STATWTO external subsystem command with the /SSR system routing command.

These external subsystem commands report the same change capture statistics but print output to different locations. The *x*EDP-STAT command prints output to the EDMMSG DD of the IMS control region, and the *x*EDP-STATWTO command prints output to the JESMSGLG DD of the IMS control region.

Note: The EDMMSG data set contains message output for PowerExchange only.

Use SDSF to view the output. The output displays the following statistics by IMS database and segment:

- Number of inserts captured
- Number of replacements captured
- Number of deletes captured

The statistics are cumulative from the time when the IMS control region started. If you need to determine change capture activity for a specific period, issue the command at the beginning of the period and again at the end of the period. Then, from the report results, subtract the earlier counts from the later counts to calculate the net capture activity for the reporting period.

Prerequisites

To use the *x*EDP-STAT and *x*EDP-STATWTO commands, ensure that the following requirements are met:

- For the /SSR command to communicate with the IMS synchronous ECCR, specify a command recognition character (CRC) when configuring the IMS external subsystem control card. To specify the CRC, edit the appropriate SSM member in the IMS PROCLIB library.
- Verify that you have the proper authority to issue the *x*EDP-STAT and *x*EDP-STATWTO commands from the IMS console. If necessary, ask the console operator to give you the required authority.
- Verify that you have the proper authority to view the command output in the EDMMSG or JESMSGLG data set. If necessary, ask the MVS system administrator to give you the required authority.

Running the Commands

Use the system routing command, /SSR , to issue the *x*EDP-STAT and *x*EDP-STATWTO commands from the IMS console.

The commands have the following syntax:

```
/SSR xEDP-STAT
/SSR xEDP-STATWTO
```

The *x* variable represents the CRC that you specified for the IMS external subsystem that is associated with the IMS synchronous ECCR. If you do not know the CRC, run the DISPLAY SUBSYS command to get it.

For more information about these commands, see the *PowerExchange Command Reference*.

Example Report Output

PowerExchange writes the following report output to the EDMMSG data set in response to the *x*EDP-STAT command. The report indicates cumulative change capture activity since the IMS control region started. The report shows the number of changes that the ECCR captured for specific segments in four IMS databases.

```
PWXEDM172853I Change Capture counts for IMS DBD DFDX01PRT
              Segment=DB#AASEG  ISRT=1        REPL=0        DLET=1
              Segment=DB#BASEG  ISRT=1        REPL=0        DLET=1
              Segment=DB#CASEG  ISRT=4        REPL=1        DLET=2
              Segment=DB#DASEG  ISRT=3        REPL=2        DLET=0
              Segment=DB#EASEG  ISRT=0        REPL=0        DLET=0
PWXEDM172853I Change Capture counts for IMS DBD FDX02PRT
              Segment=DB#AASEG  ISRT=2        REPL=1        DLET=0
              Segment=DB#BASEG  ISRT=3        REPL=1        DLET=1
              Segment=DB#CASEG  ISRT=1        REPL=0        DLET=1
              Segment=DB#BBSEG  ISRT=4        REPL=1        DLET=1
PWXEDM172853I Change Capture counts for IMS DBD FDX03PRT
              Segment=CFROOT    ISRT=5        REPL=2        DLET=1
              Segment=CFCAC01   ISRT=4        REPL=1        DLET=1
              Segment=CFCAC02   ISRT=3        REPL=0        DLET=1
PWXEDM172853I Change Capture counts for IMS DBD HSPV16
              Segment=ICROOT    ISRT=3        REPL=1        DLET=1
              Segment=POSITION  ISRT=2        REPL=2        DLET=1
              Segment=PLAYER    ISRT=4        REPL=2        DLET=0
```

To assess the change capture activity over a specific period, run the report again and then compare the reports. The second report contains the following output:

```
PWXEDM172853I Change Capture counts for IMS DBD DFDX01PRT
              Segment=DB#AASEG  ISRT=2        REPL=1        DLET=1
```

```
                Segment=DB#BASEG  ISRT=5        REPL=1        DLET=1

                Segment=DB#CASEG  ISRT=9        REPL=2        DLET=2

                Segment=DB#DASEG  ISRT=4        REPL=3        DLET=0

                Segment=DB#EASEG  ISRT=0        REPL=0        DLET=0
PWXEDM172853I Change Capture counts for IMS DBD FDX02PRT

                Segment=DB#AASEG  ISRT=4        REPL=1        DLET=1

                Segment=DB#BASEG  ISRT=5        REPL=2        DLET=1

                Segment=DB#CASEG  ISRT=3        REPL=1        DLET=1

                Segment=DB#BBSEG  ISRT=6        REPL=2        DLET=2
PWXEDM172853I Change Capture counts for IMS DBD FDX03PRT

                Segment=CFROOT    ISRT=8        REPL=3        DLET=2

                Segment=CFCAC01   ISRT=7        REPL=2        DLET=1

                Segment=CFCAC02   ISRT=6        REPL=1        DLET=1
PWXEDM172853I Change Capture counts for IMS DBD HSPV16

                Segment=ICROOT    ISRT=8        REPL=2        DLET=2

                Segment=POSITION  ISRT=4        REPL=3        DLET=1

                Segment=PLAYER    ISRT=8        REPL=3        DLET=1
```

A comparison of the reports indicates that change capture activity occurred for all segments except DB#EASEG in the DFDX01PRT database for the time interval between the first and second command issuances. To determine the specific level of capture activity, subtract the earlier counts in the first report from the later counts in the second report. For example, the following table shows the differences in change counts, or net change capture activity over the time interval, for segments in the HSPV16 database:

Segment	Later Counts	Minus Earlier Counts	Difference
ICROOT	ISRT=8 REPL=2 DLET=2	ISRT=3 REPL=1 DLET=1	ISRT=5 REPL=1 DLET=1
POSITION	ISRT=4 REPL=3 DLET=1	ISRT=2 REPL=2 DLET=1	ISRT=2 REPL=1 DLET=0
PLAYER	ISRT=8 REPL=3 DLET=1	ISRT=4 REPL=2 DLET=0	ISRT=4 REPL=1 DLET=1

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